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Adipose tissue separation device with pressure equalization

Abstract

A tissue separating device and a method of separating pure fat are provided. The tissue separating device includes a canister device including canister body having a sidewall defining a volume. A tissue retrieval port can be arranged on the canister device and arranged in fluid communication with a harvesting device for directing a fatty liposuction aspirate into the canister device. An adjustable height filtration assembly can be arranged within the canister body. A tissue harvesting port can be arranged in the sidewall of the canister body and can be in communication with a collection device to allow the tissue harvesting port to atraumatically receive a filtered pure fat. An air pressure equalizing line can be operatively mounted to the canister device to allow fluid connection between the upper vacuum chamber and the lower vacuum chamber. The filtration assembly can be movably arranged within the canister body.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATION (1) The present application claims the benefit from earlier filed U.S. Provisional Patent Application No. 63/590,721, filed Oct. 16, 2023, which is incorporated herein in its entirety by reference.

FIELD OF THE INVENTION

(1) The present teachings relate to a tissue separating device for body fluids and tissue specimens. In particular, the present teachings relate to a separation device for use with a suction source to withdraw and separate body fluids and tissue specimens from a patient in an atraumatic manner.

BACKGROUND OF THE INVENTION

(2) Autologous adipose tissue (or fatty tissue) transplantation is performed by many surgeons for various cosmetic and reconstructive procedures. Autologous fat transplantation involves retrieving adipose tissue using liposuction techniques from an area of abundance and then re-injecting the harvested adipose tissue into a different site of the same individual for cosmetic/reconstructive augmentation or enhancement purposes. Generally, prior to the re-introduction of the tissue into the patient, the adipose tissue must be processed or cleaned to maximize the chances of implant survival. Such processing is preferably accomplished while minimizing the exposure of the tissue to air as much as possible. However, the adipose cells are relatively delicate and the number of steps and length of time required to separate and process the harvested tissue prior to re-introduction into the patient contributes directly to the success of the operation and decreases the likelihood the tissue will be rejected.

(3) However, the nature of liposuction procedures preclude easy tissue isolation after initial harvest, especially on a large scale, because the volume and/or viscosity of the extracted liposuction effluent also contains unwanted components such as oil, blood and anesthetic solution. Currently, there are no standard techniques, methods, or devices that exist for the simple, large scale isolation and

refinement of liposuction-harvested adipose tissue. Although a number of specialized cannulas, needles and methods for tissue harvest and preparation exist, these techniques are tedious and inefficient. Often, the harvested fat is introduced into a centrifuge further traumatizing the fat and adding more steps to the process before the adipose tissue is re-injected back into the patient. As a result, centrifuge-free processes have been developed.

(4) Known centrifuge-free processes require a large number of steps which result in a cumbersome harvesting and re-injection process that leaves room for much improvement.

(5) Accordingly, there exists a need for an efficient, simple-to-use, low-cost to manufacture and assemble tissue collection and processing device that reduces the trauma to harvested tissue, improves the amount of useful tissue, and maintains a sterile processing environment.

SUMMARY OF THE INVENTION

(6) The present teachings provide a tissue separating device including a canister device including a canister body having a sidewall defining a volume. A tissue retrieval port can be arranged on the canister device and can be capable of being arranged in fluid communication with a harvesting device for directing a fatty liposuction aspirate into the volume of the canister device. An adjustable height filtration assembly can be arranged within the canister body and can include a filtering material separating the volume into an upper vacuum chamber and a lower vacuum chamber. A tissue harvesting port can be arranged in the sidewall of the canister body and can be capable of being arranged in communication with a collection device to allow the tissue harvesting port to atraumatically receive a filtered pure fat collected on the filtering material within the upper vacuum chamber. An air pressure equalizing line can be operatively mounted to the canister device to allow fluid connection between the upper vacuum chamber and the lower vacuum chamber. The filtration assembly can be movably arranged within the canister body such that the filtering material can be adjustable with respect to the tissue harvesting port.

(7) The present teachings also provide a tissue separating device including a canister device including a canister body having a sidewall defining a volume. An adjustable height filtration assembly can be arranged within the canister body and can include a filtering material having a top side portion and a bottom side portion. The filtering material can separate the volume into an upper vacuum chamber and a lower vacuum chamber. A tissue retrieval port can be arranged on the canister device and can be capable of being arranged in fluid communication with a harvesting device for directing a fatty liposuction aspirate into the-volume of the canister device onto the top side portion of the filtering material. An air pressure equalizing line can be operatively mounted to the canister device to allow fluid connection between the upper vacuum chamber and the lower vacuum chamber. A tissue harvesting port can be arranged in the sidewall of the canister body and can be capable of being arranged in communication with a collection device for atraumatically receiving a filtered pure fat component collected on the top side portion of the filtering material. The adjustable height filtration assembly can be movably arranged within the canister body such that the filtering material is adjustable with respect to the tissue harvesting port.

(8) The present teachings still further provide a tissue separating device including a canister device including a canister body having a sidewall defining a volume. A tissue retrieval port can be arranged on the canister device and can be capable of being arranged in fluid communication with a harvesting device for directing a fatty liposuction aspirate into the volume of the canister device. A single adjustable height filtration assembly can be arranged within the canister body and can include a filtering material separating the volume into an upper vacuum chamber and a lower vacuum chamber such that the fatty liposuction aspirate can be directed by the tissue retrieval port into the upper vacuum chamber. A tissue harvesting port can be arranged in the sidewall of the canister body and can be capable of being arranged in communication with a collection device to allow the tissue harvesting port to atraumatically receive a filtered pure fat collected on the filtering material within the upper vacuum chamber. A vacuum connector can be provided for creating a vacuum in the upper vacuum chamber. The vacuum connector and the tissue retrieval port can be

arranged on the same side of the filtering material so that vacuum is applied on the same side of the filtering material as the upper chamber. An air pressure equalizing line can be operatively mounted to the canister device to allow fluid connection between the upper vacuum chamber and the lower vacuum chamber.

(9) The present teachings further provide a method of separating pure fat from a fatty liposuction aspirate. The method includes providing a canister device defining a volume, the canister device including a tissue retrieval port, a filtration assembly including a filtering material having a top side portion and a bottom side portion and being movably arranged within the canister body, and a tissue harvesting port. The method then directs a fatty liposuction aspirate into the volume of the canister device and onto the top side portion of the filtering material of the filtration assembly. The method then separates the fatty liposuction aspirate using the filtering material into a filtered pure fat component collected on the top side portion of the filtering material and into an aqueous fluid component collected below the filtering material. The method includes moving the filtering material with respect to the tissue harvesting port into a position to place the separated pure fat collected on the top side portion of the filtering material in communication with a collection device. The method can include equalizing an air pressure between the upper vacuum chamber and the lower vacuum chamber, as well as creating a vacuum at the tissue harvesting port using the collection device to atraumatically remove the filtered pure fat from the canister body without manipulating the fat.

(10) Additional features and advantages of various embodiments will be set forth, in part, in the description that follows, and will, in part, be apparent from the description, or may be learned by the practice of various embodiments. The objectives and other advantages of various embodiments will be realized and attained by means of the elements and combinations particularly pointed out in the description herein.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 shows a side view of the tissue separating device of the present teachings;
 - (2) FIG. 2 shows a perspective exploded view of the tissue separating device of the present teachings;
 - (3) FIG. 3 shows a side cross-sectional view of the tissue separating device taken through line 3-3 of FIG. 1; and
 - (4) FIG. 4 shows a top view of the tissue separating device taken through line 4-4 of FIG. 1.
- (5) It is to be understood that both the foregoing general description and the following detailed description are exemplary and explanatory only, and are intended to provide an explanation of various embodiments of the present teachings.

DETAILED DESCRIPTION OF THE PREFERRED EMBODIMENTS

(6) Referring to at least FIG. 1, a tissue separating device **100** in accordance with a preferred embodiment of the present teachings is shown. The tissue separating device **100** can include a canister device **20** including a canister body **21** and a lid **30**. The canister device **20** can include a vacuum connector **34** and a tissue retrieval port **36** which can be arranged in the lid **30**. The vacuum connector **34** can be operatively coupled to a vacuum source **200** via tubing. The tissue retrieval port **36** can be operatively coupled to a harvesting device **300** via a liposuction cannula. A sidewall of the canister body **21** can include a fluid evacuation port **22** and a tissue harvesting port **24**. The tissue separating device **100** can also include an adjustable height filtration assembly **50** that is operable to divide the interior of the canister device **20** into an upper vacuum chamber **52** and a lower vacuum chamber **54**.

(7) In use, a quantity of fatty liposuction aspirate can be directed into the canister device **20** through

the tissue retrieval port **36** when a suction or negative pressure is applied to the sterile interior chamber of the canister device **20**. The suction pressure can be created by connecting the vacuum source **200** to the vacuum connector **34**. The fatty liposuction aspirate is directed into the upper vacuum chamber **52** and becomes filtered by the adjustable height filtration assembly **50** by way of gravity. The filtering material **62** of the filtration assembly **50** can separate pure fat from the aqueous component of the fatty liposuction aspirate. The filtered pure fat remains on the upper vacuum chamber **52** side of the filtration assembly **50** while the aqueous component of the fatty liposuction aspirate is collected at the bottom of the canister body **21** below the filtration assembly **50**. The filtering material **62** of the filtration assembly **50** can be a mesh or mesh-like material.

(8) As will be discussed in more detail below, the filtering material **62** of the filtration assembly **50** can be moved up and down the sidewall **28** of the canister body **21** and can be set at any height to facilitate the atraumatic removal of pure fat from the canister device **20** without manipulating the fat. The pure fat can be removed through the tissue harvesting port **24** by way of a collection syringe **400** or other collection mechanism. The pure fat can then be re-injected by the surgeon for fat grafting purposes. The remaining aqueous fluid component collected below the filtration assembly **50** can be separately aspirated from the canister device **20** through the fluid evacuation port **22** into a different collection syringe **500** or other collection mechanism.

(9) The tissue separating device **100** of the present teachings can be used as part of a larger tissue retrieval system. For example, the tissue separating device **100**, when operatively placed in communication with other components such as a vacuum source **200**, a waste canister (not shown), collection syringes **400**, **500**, tissue harvesting devices **300**, and others using associated conduits and tubing, can be part of a high volume, sterile, closed environment system. The system can facilitate the harvesting of aspirated, adipose tissue, and the tissue separating device **100** of the present teachings can be used to separate such tissue from other collected tumescent fluids, oils, blood, and anesthetics. As will be discussed in more detail below, pre-loading the canister device **20** with a fluid bath can be implemented to further reduce the trauma to the harvested tissue and contribute to the refining process.

(10) Referring now to FIG. **3**, the canister body **21** can include a generally cylindrically shaped sidewall **28** that can terminate in an upper rim **56** with an outwardly projecting flange **58** surrounding an open top that preferably faces upwardly when in use. The upper rim **56** forms an attachment surface for a removable lid **30** as will be described in more detail below. According to various embodiments, the removable lid **30** can include various shapes and sizes and could be formed in a hinged relationship with the canister body **21**.

(11) According to various embodiments, the canister body **21** can have a capacity from about 100 ml to about 4 liters. However, it has been found that a canister capacity of from about 1.2 liters to about 2.0 liters has been found to be suitable for most autologous adipose tissue re-injection procedures. However, these size ranges are not meant to be limiting in any manner and other suitable smaller or larger capacities as dictated by the procedure to be performed would be appreciated by one of ordinary skill in the art.

(12) As best shown in FIGS. **2** and **3**, the adjustable height filtration assembly **50** can include a filtering mesh **62** having a resilient O-ring **64** arranged about its circumference to prevent fluid from passing between the filtering mesh **62** and an interior surface of the sidewall **28** of the canister body **21**. The filtering mesh **62** can span the interior diameter of the canister body **22** and effectively form a perforated separation between the upper vacuum chamber **52** and the lower vacuum chamber **54**.

(13) The height of the filtering material **62** within the canister body **21** can be arranged to be selectively adjustable. The filtering material **62** can be secured to a movable rod **66** which extends through an aperture in the lid **30**. A top end portion of the movable rod **66** can include a handle **68** that allows a user to grasp the rod **66** and adjust the height of the filtering material **62** within the canister body **21**. The movable rod **66** and filtering material **62** can be held in place at any

particular height within the canister body **21** by way of any type of locking, gripping, or holding mechanism **70**. For example, the movable rod **66** can be secured at any height using a gasket arrangement **70** that provides a friction fit between the movable rod **66** and the lid **30**. For another example, the movable rod **66** can be arranged in threaded engagement with the lid **30** allowing concise height adjustment by turning the handle **68**.

(14) The filtering material **62** of the filtration assembly **50** can be moved up and down the canister body **21** and can be secured at any particular height within the canister body **21**. In particular, the height of the filtering material **62** of the filtration assembly **50** can be set at any location below the tissue harvesting port **24**. The adjustment of the filtering material **62** allows the surgeon to indirectly manipulate the filtered pure fat that has accumulated on the upper vacuum chamber **52** side of the filtering material **62** towards the tissue harvesting port **24**. Thus, the height adjustable filtration assembly **50** facilitates the atraumatic removal of pure fat from the canister device **20** without manipulating the fat.

(15) The perforations in the filtering material **62** are sized to allow fluids such as oils, free lipids, tumescent fluids, non-fatty tissues, blood, anesthetic fluids, and saline gathered during the harvesting process or introduced into the canister device **20** to drain from the upper vacuum chamber **52** to the lower vacuum chamber **54**. At the same time, the perforations in the filtering material **62** are sized to restrict the passage of adipose tissue. A preferable aperture size of about 1.0 mm has been determined to be suitable for the desired flow rate while avoiding clogging issues but other suitable sizes, including, but not limited to an aperture size range of about 0.2 mm to about 2.0 mm, depending on the tissue being collected, quantity desired, and desired pass through rate without undue clogging will occur to one of ordinary skill in the art.

(16) While a single filtering material **62** is shown in this embodiment, additional filtering materials could be implemented as part of the filtration assembly **50**. Depending on the vacuum strength introduced by the vacuum source **200**, some of the fatty tissue may pass through the filtering material **62** or lodge into the perforations of the material **62**. However, clogging is generally avoided since the fatty tissue tends to float and only settles onto the filtering material **62** once all other fluids in the fatty liposuction aspirate have been evacuated through the perforations and into the lower vacuum chamber **54**. While the filtering material **62** is preferably constructed of a mesh material, other suitable filtering materials could be implemented as would be appreciated by one of ordinary skill in the art.

(17) As best shown in FIGS. **1** and **2**, an air pressure equalizing line **72** can be operatively mounted to the canister body **21** such that an upper end of the air pressure equalizing line **72** can fluidically connect with the upper vacuum chamber **52** and a lower end can connect with the lower vacuum chamber **54** at a location below the filtering material **62**. Fluid flow through the air pressure equalizing line **72** can be controlled by any fluid flow control mechanism **74**, such as a stop cock valve.

(18) The valve **74** can be arranged to equalize the air pressure manually or automatically between the upper vacuum chamber **52** and the lower vacuum chamber **54**.

(19) According to various embodiments, the valve **74** can be controlled via an electrical controller **76** and pressure sensors **53**, **55** arranged in each of the upper vacuum chamber **52** and the lower vacuum chamber **54**. The electrical controller **76** can be used to automatically equalize the air pressure between the upper vacuum chamber **52** and the lower vacuum chamber **54** using pressure readings from the pressure sensors **53**, **55**.

(20) If the filtering material **62** becomes partially or fully clogged during use and the filtration assembly **50** is moved upwardly in canister body **21**, a negative pressure can form in the lower vacuum chamber **54**. This negative pressure can make it difficult for the surgeon to raise the filtering material **62** towards the tissue harvesting port **24** and could influence filtration. By opening the valve **74** and equalizing the pressure between the upper vacuum chamber **52** and the lower vacuum chamber **54** the surgeon can more readily raise the filtering material **62** within the

canister body **21** while also facilitating filtration.

(21) Covering the open top of the canister body **21** can be a removable, self-sealing lid **30**. The substantially flat lid **30** can include an outer flange **32** for mating with the outwardly projecting flange **58** of the canister body **21** to seal off the upper portion of the upper vacuum chamber **52**. The lid **30** further provides access to the vacuum connector **34** and tissue retrieval port **36** which allow admission directly into the upper vacuum chamber **52**. The lid **30** can be formed in one-piece with the canister body **21**.

(22) The fluid evacuation port **22** is arranged in fluid communication with a lower portion of the lower vacuum chamber **54**. This lower location is below the height of the filtering material **62**. An evacuation tube can be slip fit onto or into the fluid evacuation port **22** or otherwise be suitably secured to the port **22**. The fluid evacuation port **22**, a tube, and a collection syringe **500** can be used to evacuate or drain tumescent fluids, oils, bloods, anesthetics, and other unwanted bilge fluids from the lower vacuum chamber **54** under suction from the syringe or other vacuum source.

(23) The tissue harvesting port **24** is arranged in fluid communication with the upper vacuum chamber **52**. In use, the location of the tissue harvesting port **24** is above the height of the filtering material **62** to allow the filtered pure fat that has accumulated on the upper vacuum chamber **52** side of the filtering material **62** to enter the tissue harvesting port **24** and be atraumatically removed from the canister device **20**. An evacuation tube can be slip fit onto or into the tissue harvesting port **24** or otherwise be suitably secured to the port **24**. The tissue harvesting port **24**, a tube, and an adipose tissue collection syringe **400** can be used to collect the filtered pure fat from the upper vacuum chamber **52** of the tissue separating device **100**.

(24) The adipose tissue collection syringe **400** can be a conventional syringe with a hollow housing defining a collection chamber having an entry channel on one end that can attach to a tube for enabling collection of the filtered pure fat from port **24**. After the collection syringe **400** collects a quantity of filtered pure fat from the tissue separating device **100**, the syringe can be used to re-inject the filtered fat into a patient site for fat grafting or into another syringe body. Any collection syringe capacities can be implemented depending on the procedure being performed and the quantity of filtered fat is required as would be appreciated by one of ordinary skill in the art.

(25) As best shown in FIGS. **2** and **3**, the canister device **20** can be releasably secured to a holding tray **82**. For example, the canister body **21** can include tabs **26** that allow a snap-connection with the holding tray **82** via flexible tangs **84**. However, the canister device **20** can be secured to the holding tray **82** using any type of releasable connection mechanism as would be appreciated by one of ordinary skill in the art.

(26) To promote the removal of filtered pure fat from the canister device **20** into the adipose tissue collection syringe **400**, the holding tray **82** can be pivotally arranged on a base support **90**. This pivoting structure allows the surgeon to tilt the canister device **20** so that the filtered pure fat can slide on the filtering material **62** towards and through the tissue harvesting port **24**. The pivoting capability of the canister device **20** along with the height adjustability of the filtering material **62** helps the surgeon facilitate the removal of the filtered pure fat without directly manipulating the fat.

(27) The pivoting capability of the canister device **20** also helps the surgeon facilitate the removal of the aqueous fluid component of the fatty liposuction aspirate from the lower vacuum chamber **54** through the tissue retrieval port **36**.

(28) As best shown in FIG. **3**, the pivoting structure can include a pair of pivot pins **86** formed on the holding tray **82** that can engage with slots **92** formed in the base support **90** to allow the canister device **20** to be tilted in a forward and backward direction. However, the pivoting structure can include any structure that allows pivoting movement of the canister device **20** as would be appreciated by one of ordinary skill in the art. The fluid evacuation port **22** and the tissue harvesting port **24** can each be arranged on the canister body **20** in a location orthogonal to the pivot pins **86** of the holding tray **82** to allow the most efficient removal of the filtered pure fat and the aqueous fluid component of the fatty liposuction aspirate.

(29) In use, the lid **30** can be removed by the surgeon and an optional fluid bath, such as a sterile saline solution, can be poured into the canister body **21** to a desired level. It can be preferable to ensure a ratio of the fluid bath to the desired amount of aspirated adipose tissue to allow the fatty tissue to float thereby initiating the separation process. Other treatments such as antibiotics and/or sterile tissue washing and separation fluids can also be introduced at this time or later on in the process as the surgeon determines. Such refining fluids can be introduced to wash, separate, and treat the aspirated adipose tissue.

(30) At this point, the vacuum connector **34** can be placed in communication with the vacuum source **200** and the tissue retrieval port **36** can be placed in communication with the harvesting device **300**. The tissue harvesting port **24** can be coupled to the adipose tissue collection syringe **400** and can be closed off by a pinch clamp. Similarly, the fluid evacuation port **22** can be coupled to a collection syringe **500** or other collection mechanism and can be closed off by a pinch clamp.

(31) The vacuum source **200** can be turned on by the surgeon and the pump allowed to build to a suitable vacuum pressure to perform the fatty tissue harvesting process with the harvesting device **300** using conventional harvesting techniques. With the vacuum source **200** activated, a vacuum pressure relative to the ambient pressure can be generated and maintained in the upper vacuum chamber **52**. A vacuum pressure range of 15-20 inches Hg has been found suitable to maintain a sufficient suction force in the upper vacuum chamber **52** to provide sufficient suction to the harvesting device **300** for aspirating adipose tissue from a patient and into the canister device **20**.

(32) Using conventional liposuction techniques, the cannula of the harvesting device **300** can be inserted into a patient site and a quantity of adipose tissue harvested under vacuum. The fatty liposuction aspirate will travel through the fat harvesting tube and is directed through the tissue retrieval port **36** and fall into the fluid bath or onto the filtering material **62** in the upper vacuum chamber **40**. As commonly occurs during the adipose tissue aspiration process, other fluids and tissues such as blood, anesthetic, collagen strands, infiltrated tumescent fluids, oils, and other non-fatty tissues and fluids may be aspirated along with the fatty tissue. It will be appreciated that the process of separating the fatty tissue from other fluids and non-fatty tissue and fluids begins immediately.

(33) While the fatty tissue can initially sink as it is introduced into the fluid bath, the more buoyant fatty tissue tends to rise within the fluid bath and will ultimately float at or near the upper water line of the fluid bath. It will be appreciated that pre-loading the canister device **20** with a fluid bath can cushion the entry of the fatty tissue into the canister thus reducing the trauma to the tissue as well as irrigating the tissue. As the fluid bath mixes with the aspirated, adipose tissue, it becomes bilge fluid that can later be evacuated further on in the procedure.

(34) Once enough aspirated, adipose tissue is collected as determined by the surgeon to complete the transplantation or re-injection procedure, the vacuum source **200** can be left on at the same pressure or at a reduced pressure. However, it is preferable to turn off the vacuum source **200** while the fatty liposuction aspirate sits on the filtering material **62** and/or in the fluid bath for a short time interval to allow the separation process to proceed.

(35) During the separation process, the filtered fatty tissue is prevented from entering the lower vacuum chamber **54** due to the size of the perforations in the filtering material **62**. In addition, as the fatty tissue tends to float on the fluid bath, it is unlikely to block the perforations of the filtering material **62**.

(36) The fluid, including both saline bath and non-fatty tissues and other fluids collected from the patient during the harvesting procedure, passes from the upper vacuum chamber **52** and through the perforations in the filtering material **62**, thereby accumulating in the lower vacuum chamber **54**. The fluid that has accumulated in the lower vacuum chamber **54** can be evacuated through the fluid evacuation port **22** via a tube and into a collection syringe **500** or other collection mechanism.

(37) The filtered pure fat that settles onto the filtering material **62** is separated and rinsed all within the sterile closed environment of the canister device **20**. The filtered pure fat is heavily

concentrated and ready for re-introduction into the patient as previously discussed above. However, after these washing and separation steps, other treatment additives can be added and repeated one or more times to further refine the fatty tissue.

(38) The surgeon can open the pinch clamp downstream from the tissue harvesting port **24**, tilt the canister device **20**, and draw back the syringe plunger on the adipose tissue collection syringe **400**. This places the collection syringe **400** in more efficient communication with the filtered pure fat that has collected on top of the filtering material **62**. The tilting of the canister device **20** gently slopes the filtering material **62** to help provide a fat retrieval chute that directs the filtered pure fat to the collection syringe **400**.

(39) If more than one collection syringe is needed, the primary syringe can be uncoupled and additional syringes as needed may be coupled to the tubing end and the collection process repeated to load one or more syringes with filtered pure fat for re-injection.

(40) The collection syringe **400** loaded with filtered pure fat can then be coupled to a desired cannula by the surgeon, and the filtered pure fat re-injected into the patient site using conventional autologous fatty tissue re-injection techniques.

(41) The tissue separating device **100** of the present teachings allows the collection of filtered pure fat from aspirated adipose tissue without opening the canister device **20** and manipulating the fat. The tissue separating device **100** also minimizes the number of steps required by a surgeon to harvest, refine, and collect the filtered pure fat prior to re-introduction into the patient site. Trauma to the tissue, commonly caused by the use of a centrifuge, rougher manipulation of the tissue and the tissue container as commonly occurs when directing the tissue against a solid surface, and extending the time of the tissue outside the patient is significantly reduced, all while maintaining a sterile environment. The tissue separating device **100** of the present teachings can be easily integrated with conventional off the shelf waste canisters, vacuum sources, syringes, and harvesting devices.

(42) Those skilled in the art can appreciate from the foregoing description that the present teachings can be implemented in a variety of forms. Therefore, while these teachings have been described in connection with particular embodiments and examples thereof, the true scope of the present teachings should not be so limited. Various changes and modifications may be made without departing from the scope of the teachings herein.

Claims

1. A tissue separating device comprising: a canister device including canister body having a sidewall defining a volume; a tissue retrieval port arranged on the canister device and capable of being arranged in fluid communication with a harvesting device for directing a fatty liposuction aspirate into the volume of the canister device; a single adjustable height filtration assembly arranged within the canister body and including a filtering material separating the volume into an upper vacuum chamber and a lower vacuum chamber such that the fatty liposuction aspirate is directed by the tissue retrieval port into the upper vacuum chamber; a tissue harvesting port arranged in the sidewall of the canister body and capable of being arranged in communication with a collection device to allow the tissue harvesting port to atraumatically receive a filtered pure fat collected on the filtering material within the upper vacuum chamber; and an air pressure equalizing line operatively mounted to the canister device to allow fluid connection between the upper vacuum chamber and the lower vacuum chamber; wherein the filtration assembly is movably arranged within the canister body such that the filtering material is adjustable with respect to the tissue harvesting port as air pressure is equalized between the upper vacuum chamber and the lower vacuum chamber via the air pressure equalization line to alleviate a negative pressure in the lower vacuum chamber when the filtering material becomes clogged.

2. The tissue separating device of claim 1, further including a valve arranged in the air pressure

equalizing line capable of selectively controlling the fluid connection between the upper vacuum chamber and the lower vacuum chamber.

3. The tissue separating device of claim 2, wherein the valve is a stop cock.

4. The tissue separating device of claim 2, further including an electrical controller capable of selectively controlling the valve arranged in the air pressure equalizing line to control the fluid connection between the upper vacuum chamber and the lower vacuum chamber.

5. The tissue separating device of claim 4, wherein the electrical controller selectively controls the valve based on pressure readings from at least one of the upper vacuum chamber and the lower vacuum chamber.

6. A tissue separating device comprising: a canister device including canister body having a sidewall defining a volume; an adjustable height filtration assembly arranged within the canister body and including a filtering material having a top side portion and a bottom side portion, the filtering material separating the volume into an upper vacuum chamber and a lower vacuum chamber; a tissue retrieval port arranged on the canister device and capable of being arranged in fluid communication with a harvesting device for directing a fatty liposuction aspirate into the volume of the canister device onto the top side portion of the filtering material; an air pressure equalizing line operatively mounted to the canister device to allow fluid connection between the upper vacuum chamber and the lower vacuum chamber; and a tissue harvesting port arranged in the sidewall of the canister body and capable of being arranged in communication with a collection device for atraumatically receiving a filtered pure fat component collected on the top side portion of the filtering material; wherein the adjustable height filtration assembly is movably arranged within the canister body such that the filtering material is adjustable with respect to the tissue harvesting port as air pressure is equalized between the upper vacuum chamber and the lower vacuum chamber via the air pressure equalization line to alleviate a negative pressure in the lower vacuum chamber when the filtering material becomes clogged.

7. The tissue separating device of claim 6, further including a valve arranged in the air pressure equalizing line capable of selectively controlling the fluid connection between the upper vacuum chamber and the lower vacuum chamber.

8. The tissue separating device of claim 7, wherein the valve is a stop cock.

9. The tissue separating device of claim 7, further including an electrical controller capable of selectively controlling the valve arranged in the air pressure equalizing line to control the fluid connection between the upper vacuum chamber and the lower vacuum chamber.

10. The tissue separating device of claim 9, wherein the electrical controller selectively controls the valve based on pressure readings from at least one of the upper vacuum chamber and the lower vacuum chamber.

11. A tissue separating device comprising: a canister device including canister body having a sidewall defining a volume; a tissue retrieval port arranged on the canister device and capable of being arranged in fluid communication with a harvesting device for directing a fatty liposuction aspirate into the volume of the canister device; a single adjustable height filtration assembly arranged within the canister body and including a filtering material separating the volume into an upper vacuum chamber and a lower vacuum chamber such that the fatty liposuction aspirate is directed by the tissue retrieval port into the upper vacuum chamber; a tissue harvesting port arranged in the sidewall of the canister body and capable of being arranged in communication with a collection device to allow the tissue harvesting port to atraumatically receive a filtered pure fat collected on the filtering material within the upper vacuum chamber; a vacuum connector for creating a vacuum in the upper vacuum chamber, wherein the vacuum connector and the tissue retrieval port are arranged on the same side of the filtering material so that vacuum is applied on the same side of the filtering material as the upper chamber; and an air pressure equalizing line operatively mounted to the canister device to allow fluid connection between the upper vacuum chamber and the lower vacuum chamber as a height of the filtering material is adjusted with respect

to the tissue harvesting port to alleviate a negative pressure in the lower vacuum chamber when the filtering material becomes clogged.

12. The tissue separating device of claim 11, further including a valve arranged in the air pressure equalizing line capable of selectively controlling the fluid connection between the upper vacuum chamber and the lower vacuum chamber.

13. The tissue separating device of claim 12, wherein the valve is a stop cock.

14. The tissue separating device of claim 12, further including an electrical controller capable of selectively controlling the valve arranged in the air pressure equalizing line to control the fluid connection between the upper vacuum chamber and the lower vacuum chamber.

15. The tissue separating device of claim 14, wherein the electrical controller selectively controls the valve based on pressure readings from at least one of the upper vacuum chamber and the lower vacuum chamber.

16. A method of separating pure fat from a fatty liposuction aspirate comprising: providing a canister device defining a volume, the canister device including a tissue retrieval port, a filtration assembly including a filtering material having a top side portion and a bottom side portion and being movably arranged within the canister body, and a tissue harvesting port; directing a fatty liposuction aspirate into the volume of the canister device and onto the top side portion of the filtering material of the filtration assembly; separating the fatty liposuction aspirate using the filtering material into a filtered pure fat component collected on the top side portion of the filtering material and into an aqueous fluid component collected below the filtering material; moving the filtering material with respect to the tissue harvesting port into a position to place the separated pure fat collected on the top side portion of the filtering material in communication with a collection device; equalizing an air pressure between the upper vacuum chamber and the lower vacuum chamber as the filtering material is moved with respect to the tissue harvesting port to alleviate a negative pressure in the lower vacuum chamber when the filtering material becomes clogged; and creating a vacuum at the tissue harvesting port using the collection device to atraumatically remove the filtered pure fat from the canister body without manipulating the fat.

17. The method of separating pure fat from a fatty liposuction aspirate of claim 16, further comprising tilting the canister device to promote the removal of the filtered pure fat from the canister body.

18. The method of separating pure fat from a fatty liposuction aspirate of claim 16, wherein equalizing an air pressure between the upper vacuum chamber and the lower vacuum chamber includes a valve capable of selectively controlling the air pressure between the upper vacuum chamber and the lower vacuum chamber.

19. The method of separating pure fat from a fatty liposuction aspirate of claim 18, wherein the valve is a stop cock.

20. The method of separating pure fat from a fatty liposuction aspirate of claim 18, further including an electrical controller capable of selectively controlling the valve.
